

Project Part 6: Predefined queries with output

1. Calculate the total Fines by Member

```
SELECT

    bt.ClientID,

    c.Name                AS MemberName,

    SUM(

        GREATEST(

            DATEDIFF(

                COALESCE(bt.ReturnDate, CURRENT_DATE),

                bt.DueDate

            ), 0

        )

        * m.DailyFineRate

    )                    AS TotalFine

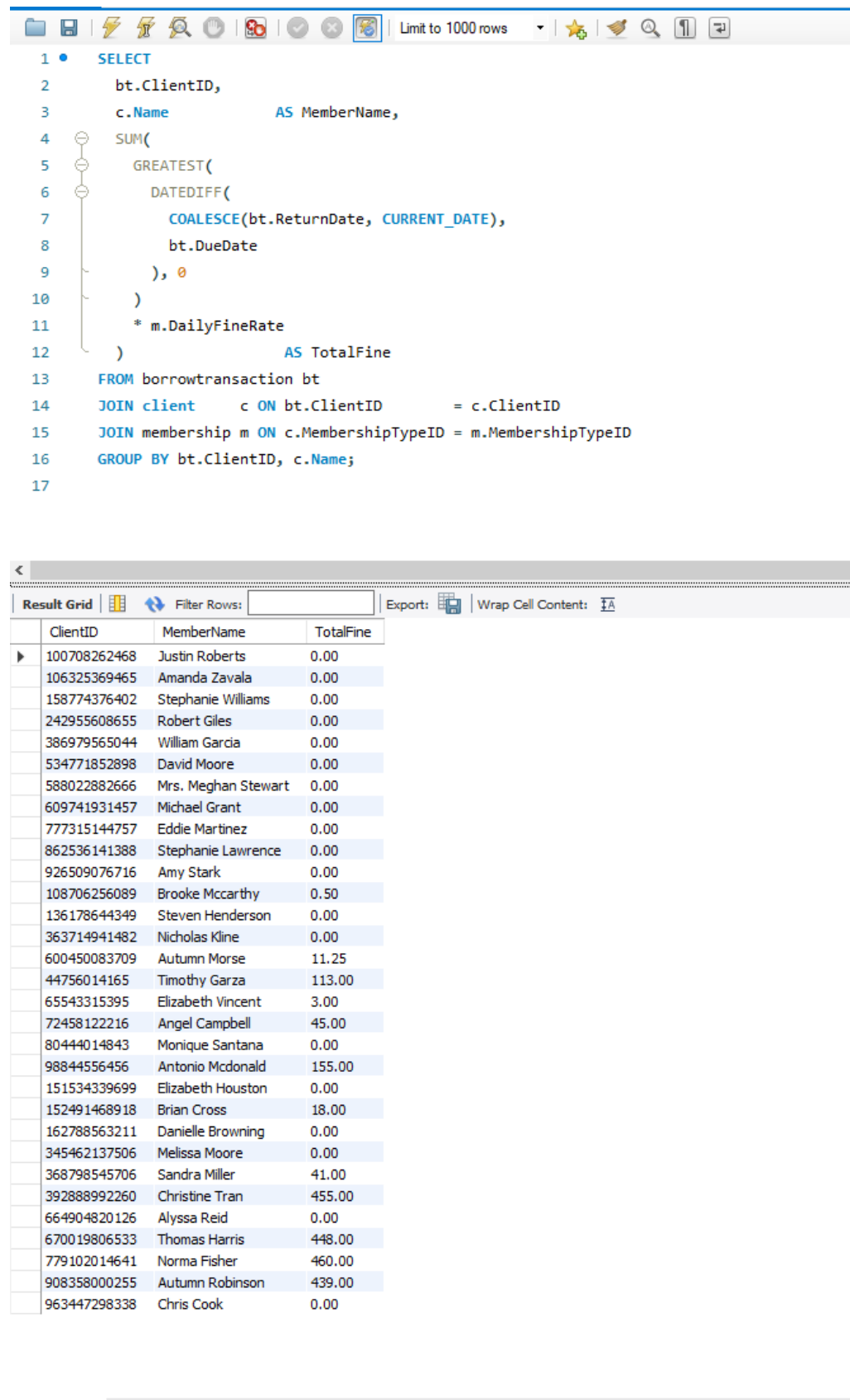
FROM borrowtransaction bt

JOIN client      c ON bt.ClientID      = c.ClientID

JOIN membership m ON c.MembershipTypeID = m.MembershipTypeID

GROUP BY bt.ClientID, c.Name;
```

Output Screenshot:



The screenshot displays a SQL query in a query editor and its corresponding result set in a table.

SQL Query:

```
1 SELECT
2     bt.ClientID,
3     c.Name AS MemberName,
4     SUM(
5         GREATEST(
6             DATEDIFF(
7                 COALESCE(bt.ReturnDate, CURRENT_DATE),
8                 bt.DueDate
9             ), 0
10        )
11        * m.DailyFineRate
12    ) AS TotalFine
13 FROM borrowtransaction bt
14 JOIN client c ON bt.ClientID = c.ClientID
15 JOIN membership m ON c.MembershipTypeID = m.MembershipTypeID
16 GROUP BY bt.ClientID, c.Name;
```

Result Grid:

ClientID	MemberName	TotalFine
100708262468	Justin Roberts	0.00
106325369465	Amanda Zavala	0.00
158774376402	Stephanie Williams	0.00
242955608655	Robert Giles	0.00
386979565044	William Garcia	0.00
534771852898	David Moore	0.00
588022882666	Mrs. Meghan Stewart	0.00
609741931457	Michael Grant	0.00
777315144757	Eddie Martinez	0.00
862536141388	Stephanie Lawrence	0.00
926509076716	Amy Stark	0.00
108706256089	Brooke Mccarthy	0.50
136178644349	Steven Henderson	0.00
363714941482	Nicholas Kline	0.00
600450083709	Autumn Morse	11.25
44756014165	Timothy Garza	113.00
65543315395	Elizabeth Vincent	3.00
72458122216	Angel Campbell	45.00
80444014843	Monique Santana	0.00
98844556456	Antonio Mcdonald	155.00
151534339699	Elizabeth Houston	0.00
152491468918	Brian Cross	18.00
162788563211	Danielle Browning	0.00
345462137506	Melissa Moore	0.00
368798545706	Sandra Miller	41.00
392888992260	Christine Tran	455.00
664904820126	Alyssa Reid	0.00
670019806533	Thomas Harris	448.00
779102014641	Norma Fisher	460.00
908358000255	Autumn Robinson	439.00
963447298338	Chris Cook	0.00

2. Display a list of all available books

```
SELECT
```

```
    b.ISBN,
```

```
    b.Title,
```

```
    b.Author,
```

```
    b.Genre
```

```
FROM book b
```

```
JOIN copy cp ON b.CopyID = cp.CopyID
```

```
WHERE cp.AvailabilityStatus = 'Available'
```

```
    AND b.Genre = 'Mystery';
```

Output Screenshot:

```
1 • SELECT
2     b.ISBN,
3     b.Title,
4     b.Author,
5     b.Genre
6 FROM book b
7 JOIN copy cp ON b.CopyID = cp.CopyID
8 WHERE cp.AvailabilityStatus = 'Available'
9     AND b.Genre = 'Mystery';
```

<				
Result Grid				
Filter Rows:				
Export:  Write				
	ISBN	Title	Author	Genre
▶	1023832279531	Twilight Covenant	Orion Black	Mystery
	MST0001	Mystery Dawn	Alice Chen	Mystery
	MST0002	Mystery Dusk	Bob Li	Mystery
	MST0005	Mystery Echoes	Eve Zhang	Mystery
	MST0006	Mystery Winds	Frank Gao	Mystery
	MST0008	Mystery Codes	Henry Wu	Mystery

3. Identify the members who have borrowed the most books in a particular genre in last year

SELECT

bt.ClientID,

c.Name AS MemberName,

COUNT(*) AS BorrowCount

FROM borrowtransaction bt

JOIN client c ON bt.ClientID = c.ClientID

JOIN book b ON bt.CopyID = b.CopyID

WHERE b.Genre = 'Mystery'

AND bt.CheckoutDate >= DATE_SUB(CURRENT_DATE, INTERVAL 1
YEAR)

GROUP BY bt.ClientID, c.Name

ORDER BY BorrowCount DESC

LIMIT 1;

Output Screenshot:

```
1 • SELECT
2     bt.ClientID,
3     c.Name      AS MemberName,
4     COUNT(*)    AS BorrowCount
5 FROM borrowtransaction bt
6 JOIN client c ON bt.ClientID = c.ClientID
7 JOIN book b ON bt.CopyID = b.CopyID
8 WHERE b.Genre = 'Mystery'
9     AND bt.CheckoutDate >= DATE_SUB(CURRENT_DATE, INTERVAL 1 YEAR)
10 GROUP BY bt.ClientID, c.Name
11 ORDER BY BorrowCount DESC
12 LIMIT 1;
13
```

<

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content:  | Fetch rows:

	ClientID	MemberName	BorrowCount
▶	100708262468	Justin Roberts	2

4. Generate a report of all books due within the next week, sorted by due date

SELECT

b.ISBN,

b.Title,

bt.DueDate,

c.Name AS Borrower

FROM borrowtransaction bt

JOIN client c ON bt.ClientID = c.ClientID

JOIN book b ON bt.CopyID = b.CopyID

WHERE bt.ReturnDate IS NULL

AND bt.DueDate BETWEEN CURRENT_DATE

AND DATE_ADD(CURRENT_DATE, INTERVAL 7
DAY)

ORDER BY bt.DueDate;

Output Screenshot:

```
1 • SELECT
2     b.ISBN,
3     b.Title,
4     bt.DueDate,
5     c.Name AS Borrower
6 FROM borrowtransaction bt
7 JOIN client c ON bt.ClientID = c.ClientID
8 JOIN book b ON bt.CopyID = b.CopyID
9 WHERE bt.ReturnDate IS NULL
10    AND bt.DueDate BETWEEN CURRENT_DATE
11                        AND DATE_ADD(CURRENT_DATE, INTERVAL 7 DAY)
12 ORDER BY bt.DueDate;
```

< **Result Grid** |   Filter Rows: | Export:  | Wrap Cell Content: 

	ISBN	Title	DueDate	Borrower
▶	8049194392752	The Forgotten Kingdom	2025-05-07	Antonio Mcdonald
	8049194392752	The Forgotten Kingdom	2025-05-08	Justin Roberts
	9158736461602	Dreams of Glass	2025-05-08	Elizabeth Vincent

5. List all members who currently have at least one overdue book, along with the titles of the overdue books.

```
SELECT
```

```
    bt.ClientID,
```

```
    c.Name           AS MemberName,
```

```
    GROUP_CONCAT(b.Title SEPARATOR ', ') AS OverdueTitles
```

```
FROM borrowtransaction bt
```

```
JOIN client c      ON bt.ClientID = c.ClientID
```

```
JOIN book  b       ON bt.CopyID    = b.CopyID
```

```
WHERE bt.ReturnDate IS NULL
```

```
    AND bt.DueDate < CURRENT_DATE
```

```
GROUP BY bt.ClientID, c.Name;
```

Output Screenshot:

```
1 • SELECT
2     bt.ClientID,
3     c.Name          AS MemberName,
4     GROUP_CONCAT(b.Title SEPARATOR ', ') AS OverdueTitles
5 FROM borrowtransaction bt
6 JOIN client c      ON bt.ClientID = c.ClientID
7 JOIN book  b       ON bt.CopyID   = b.CopyID
8 WHERE bt.ReturnDate IS NULL
9       AND bt.DueDate < CURRENT_DATE
10 GROUP BY bt.ClientID, c.Name;
11
```

<			
Result Grid			
Filter Rows:			
Export:			
Wrap Cell Content:			
	ClientID	MemberName	OverdueTitles
▶	72458122216	Angel Campbell	Echoes of Tomorrow
	98844556456	Antonio Mcdonald	The Crystal Code, Dreams of Glass
	242955608655	Robert Giles	Shadows of Eternity
	368798545706	Sandra Miller	The Forgotten Kingdom
	600450083709	Autumn Morse	Whispers in the Wind
	779102014641	Norma Fisher	Beneath the Iron Sky

6. Calculate the average number of days members borrow books for a specific genre.

```
SELECT
```

```
    b.Genre,
```

```
    AVG(
```

```
        DATEDIFF(
```

```
            COALESCE(bt.ReturnDate, CURRENT_DATE),
```

```
            bt.CheckoutDate
```

```
        )
```

```
    ) AS AvgBorrowDays
```

```
FROM borrowtransaction bt
```

```
JOIN book b      ON bt.CopyID = b.CopyID
```

```
WHERE b.Genre = 'Mystery'
```

```
GROUP BY b.Genre;
```

Output Screenshot:

```
1 • SELECT
2     b.Genre,
3     AVG(
4     DATEDIFF(
5         COALESCE(bt.ReturnDate, CURRENT_DATE),
6         bt.CheckoutDate
7     )
8     ) AS AvgBorrowDays
9 FROM borrowtransaction bt
10 JOIN book b ON bt.CopyID = b.CopyID
11 WHERE b.Genre = 'Mystery'
12 GROUP BY b.Genre;
13
```

<

Result Grid | | Filter Rows: | Export: | Wrap Cell Content:

	Genre	AvgBorrowDays
▶	Mystery	16.4000

7. Determine the author whose books have been borrowed the most in the last month.

```
SELECT
```

```
    b.Author,
```

```
    COUNT(*)          AS BorrowCount
```

```
FROM borrowtransaction bt
```

```
JOIN book b          ON bt.CopyID = b.CopyID
```

```
WHERE bt.CheckoutDate >= DATE_SUB(CURRENT_DATE, INTERVAL 1  
MONTH)
```

```
GROUP BY b.Author
```

```
ORDER BY BorrowCount DESC
```

```
LIMIT 1;
```

Output Screenshot:

```
1 • SELECT
2     b.Author,
3     COUNT(*)          AS BorrowCount
4 FROM borrowtransaction bt
5 JOIN book b          ON bt.CopyID = b.CopyID
6 WHERE bt.CheckoutDate >= DATE_SUB(CURRENT_DATE, INTERVAL 1 MONTH)
7 GROUP BY b.Author
8 ORDER BY BorrowCount DESC
9 LIMIT 1;
10
```

<		
Result Grid		Filter Rows:
Export:  Wrap Cell Content:  Fetch rows: 		
	Author	BorrowCount
▶	Damien Hale	3